Agentic Traffic Management System

Project Report & Development Plan

1. Background

The original research studied the relationship between traffic congestion and vehicular emissions. While most studies assume that congestion and emissions always increase together, the research showed that this is not always true. This phenomenon is known as Emission–Congestion Decoupling.

2. What Has Already Been Completed?

Speed Calculation – Average speed was calculated using travel time and distance.

Congestion Index – Measures how much slower traffic moves compared to free-flow traffic.

Congestion Index = (Free Flow Speed – Average Speed) / Free Flow Speed

Emission Calculation – Vehicle emissions were estimated using a speed-based emission model.

CO■ $= a + bV + cV^2$

Decoupling Analysis – Determines whether congestion and emissions move together or independently.

XGBoost Prediction – Predicts whether a traffic scenario exhibits decoupling behaviour.

3. Reusable Components from Existing Research

Component	Future Use
Congestion Formula	Congestion Agent
Emission Formula	Emission Agent
Decoupling Formula	Decoupling Agent
XGBoost Model	Prediction Agent
Traffic Categories	SUMO Scenarios
Synthetic Traffic Logic	Simulation Baseline

4. What Needs To Be Built?

• SUMO Integration • LangGraph Workflow • Congestion Propagation Agent • Decision Agent • Action Agent • Signal Control Logic

5. Proposed Workflow

SUMO Simulation → Congestion Agent → Propagation Agent → Emission Agent → Prediction Agent (XGBoost) → Decision Agent → Action Agent → SUMO Simulation

6. Agent Descriptions

Congestion Monitoring Agent: Determines congestion severity using speed, delay and queue length.

Congestion Propagation Agent: Predicts whether congestion will spread to nearby intersections.

Emission Impact Agent: Calculates environmental impact using emission formulas.

Prediction Agent: Uses XGBoost to predict decoupling probability.

Decision Agent: Prioritizes intersections for intervention.

Action Agent: Applies traffic signal interventions.

7. The Two Major Decisions

Decision 1 – Congestion Propagation Prediction

Identify intersections that are likely to create future congestion problems. Output: Propagation Risk Score.

Decision 2 – Intervention Prioritization

Select the intersection that should receive intervention first using congestion, propagation risk, emission impact and decoupling probability.

8. Priority Logic

Priority Score = 0.30 × Congestion Score + 0.40 × Propagation Risk + 0.20 × Emission Impact + 0.10 × Decoupling Probability

9. Project Phases

Phase 1 – Research & Planning

Understand the paper, review formulas and XGBoost, design agents, design the SUMO network and define inputs/outputs.

Phase 2 – Development

Build the SUMO simulation and develop the Congestion, Propagation, Emission, Prediction and Decision agents.

Phase 3 – Integration & Evaluation

Integrate LangGraph, implement the Action Agent, connect SUMO with the agents, test scenarios and prepare the demo.

10. Final Vision

The final system will predict congestion propagation, analyze emissions, predict decoupling behaviour, prioritize intersections, apply adaptive traffic signal control and evaluate improvements inside a SUMO simulation. This extends the original research from a prediction model into an Agentic AI decision-making framework.